Subject: Re: Tensor networks and spin networks
From: Takato Mori <moritaka@post.kek.jp>
Date: 02/03/21, 12:51 am
To: Deepak Vaid <dvaid79@gmail.com>

Also, I think writing in a way (10) is quite parallel to the PEPS construction. Although PEPS is mathematically completely equivalent to TNS, it provides a useful insights; The internal indices or bonds in TNs are essentially entanglement between sites. I have once tried to formulate sitewise entanglement based on PEPS and generalize it to an arbitrary TN, but have yet not completed. I'm still interested in that direction as well (as it seems essentially same as the BH entropy counting in spin networks).

Taka

> 2021/03/02 4:10、Takato Mori <moritaka@post.kek.jp>のメール:
>
> Hi Deepak,
>
> Sorry for the late reply. I've been busy for last few weeks. Unfortunately I've only skimmed your paper until Sec. 4. So far I think it is really well written and the correspondence between spin networks and tensor networks is clear to me. I guess the review part of the tensor network is mostly based on [32]. I think the expression of TNS (10) is not much familiar to people in the community of tensor networks as we usually group the adjacent edge tensors with a vertex tensor at each site using the gauge redundancies of TN like we do in MPS. I understand (10) is more convenient to relate it to the spin network state (25). It might be more logically clear if you discuss the gauge redundancies of TN before presenting TNS states in the way of (10).
>
> Besides, I think you should discuss boundary operators, which controls boundary conditions on tensor networks. It is possible to express an open boundary condition as a boundary covector and the periodic b.c. as a trivial boundary operator. Furthermore, we can consider a probabilistic mixture of boundary conditions and this can be expressed with a general boundary operator. For a variational purpose, it is important to include this boundary operator as well to the variational parameters. In the context of spin networks, I guess boundary operators reflects the topology and the boundary conditions of the emergent spacetime (but I am not sure.) Though, mathematically it is just edge states with no physical indices connecting open edges non-locally. I think it might be interesting to consider some nontrivial boundary operator and its consequence in spin networks.
>
> As for the boundary operators, I couldn't find good literatures somewhat. I myself remember leaned it in a Japanese textbook and a colleague H. Matsueda. Choices of matrices in (2.31) in (https://link.springer.com/chapter/10.1007/978-3-030-34489-4_2) is close to what I meant. Also, I wrote a review about it before so I attached it. Please do not redistribute. I hope you find it helpful. For the mixed boundary condition, please refer to Sec. 2.4.1 of the attached PDF.
>
> Best regards,
> <master.pdf>
>
> Taka

> 2021/03/02 2:59、Deepak Vaid <dvaid79@gmail.com>のメール:
>
> Hi Takato,
>
> Did you get a chance to look at my paper?
>
> Best,
>
> Deepak
>
> On 18/02/21 10:10 am, Takato Mori wrote:
>
>> Hi Deepak,
>>
>> Thanks for the paper. I will take a look at it!
>> (It might take a several days or a week to read through as I am a bit busy now and these days. Sorry in advance about that.)
>>
>> Takato
>>
>>> 2021/02/18 4:51、Deepak Vaid <dvaid79@gmail.com>のメール:
>>>
>>> Hi Takato,
>>>
>>> Sorry it took me so long to send you this manuscript. I was full of confidence about it at one time, but I think letting it sit untouched for too long has allowed doubts to creep up on me.
>>>
>>> You know enough about both tensor networks, spin networks and quantum gravity to be able to form an opinion about this paper. It is incomplete in parts. It would be nice if I could get some validation regarding the core ideas. That would give me the confidence to proceed further. In any case I look forward to your comments, whether positive or negative.
>>>
>>> Hope this finds you well,
>>>
>>> Best,
>>>
>>> Deepak
>>> <timesarrow.pdf>
>
> --
> Deepak Vaid
>
> https://www.quantumofgravity.com/blog